

7. The LCM and GCF of two positive integers $x+1$ and $x-1$ are 7 and 32 respectively. Find the sum of these integers.
- A. 36 B. 48 C. 60 D. 26
8. The solution set of the system $\begin{cases} \frac{3}{2x} - \frac{1}{y} = 4 \\ \frac{10}{x} - \frac{6}{y} = 1 \end{cases}$ is :
- A. $\{(1,0)\}$ B. $\{(2,3)\}$ C. $\{(-1,0)\}$ D. $\{(3,0)\}$
9. The solution set of the equation $|3x - 8| = 4 - x$ is
- A. $\{-1,3\}$ B. $\{-4,3\}$ C. $\{2,3\}$ D. $\{-4, -3\}$
10. If $\sqrt{(2x-1)^2} = 3-x$ then the value of x is
- A. $\{\frac{4}{3}, 4\}$ B. $\{\frac{4}{3}\}$ C. $\{2\}$ D. $\{\frac{2}{3}, 2\}$
11. If the order pair $(2,-3)$ is the solution of the system $\begin{cases} ax + by = 10 \\ 2ax - (b+1)y = 1 \end{cases}$ then the value of a is
- A. 1 B. -5 C. 3 D. -4
12. If θ is in standard position and its terminal side having a point $(\frac{1}{2}, -\frac{\sqrt{3}}{2})$, then find $\tan\theta, \sec\theta$, It is equal to:
- A. $\frac{-6}{\sqrt{3}}$ C. $\frac{-\sqrt{3}}{2}$
 B. $-\frac{1}{\sqrt{3}}$ D. $\frac{\sqrt{3}}{2}$
13. If $\sin\theta = -\frac{3}{4}$ and θ is not in the IV quadrant, then which one of the following is not true ?
- A. $\tan\theta = \frac{3}{\sqrt{7}}$ C. $\sec\theta \cdot \csc\theta = \frac{3}{\sqrt{7}}$
 B. $\cos\theta = \frac{\sqrt{7}}{4}$ D. $\cot\theta = \frac{\sqrt{7}}{3}$
14. If $\vec{A} = (1, -2)$, $\vec{B} = (-1, 2)$ and $\vec{C} = (-3, -4)$ are vectors, then which of the following is true?
- A. The length of $\vec{A} + \vec{C} - \vec{B}$ is 65. C. $3\vec{C} + \vec{B} - \vec{A} = (11, -8)$
 B. $-2\vec{C} - 2\vec{B} = (-4, 4)$. D. $|\vec{A}| \neq |\vec{B}|$
15. Let $\vec{A} = (3, 4)$, $\vec{B} = (2, -3)$, then $3\vec{A} - 4\vec{B}$ is equal to:
- A. (17, -24) C. (1, 0)
 B. (1, 24) D. (8, 10)
16. The magnitude of a vector with initial point $(-4, -2)$ and terminal point $(-3, 1)$ is:
- A. $\sqrt{10}$ unit C. 5 unit
 B. $\sqrt{503}$ unit D. $\sqrt{5}$ unit

17. There are 2 green, 3 yellow and 1 red balls of the same size, shape. If two balls are drawn at random, what is the probability that they are from green, yellow or red of different color at the same time appear?

A. $\frac{2}{5}$
B. $\frac{1}{3}$

C. $\frac{1}{15}$
D. $\frac{11}{15}$

18. In the following table there is a frequency distribution of a population function X, find the median of population :-

X	0	1	2	3	6
F	1	5	5	3	1

A. 1

C. 3

B. 2

D. 2.5

19. Find the population variance in the table given above question

A. $\sqrt{\frac{28}{17}}$

C. $\frac{28}{15}$

B. 2

D. $\frac{17}{9}$

20. If the circumference of a circular region is 16π cm, then what is the area of the region?

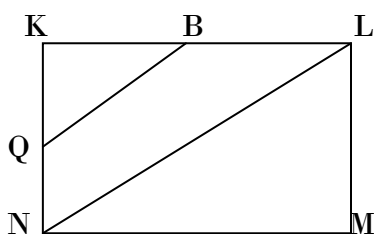
A. $64\pi \text{ cm}^2$

C. $16\pi \text{ cm}^2$

B. $32\pi \text{ cm}^2$

D. $48\pi \text{ cm}^2$

21. In the figure below, KLMN is a square of area 20 cm^2 and B and Q are mid points of sides KL and KN respectively. What is the area of $\triangle BQNL$?



A. 10 cm^2

B. $\sqrt{5} \text{ cm}^2$

C. 5 cm^2

D. 7.5 cm^2

22. The length of an altitude and radius of base of a right circular cylinder are equal. If the lateral surface area of the cylinder is $72\pi \text{ cm}^2$, then the length of the altitude is

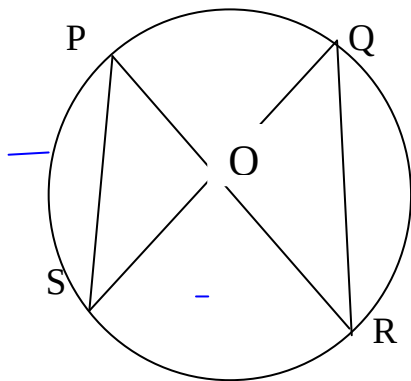
A. 6cm

C. 216 cm

B. 36cm

D. $6\sqrt{2} \text{ cm}$

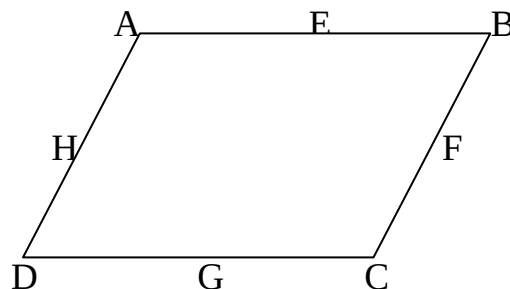
23. In the figure below O is centre of the circle, If $m(\text{arc QR}) = 40^\circ$ and $m\angle PSQ = (3y + 25)^\circ$ and $m\angle PRQ = (2y + 30)^\circ$ then $m\angle PQR$ is



- A. 80°
- B. 85°
- C. 120°
- D. 90°

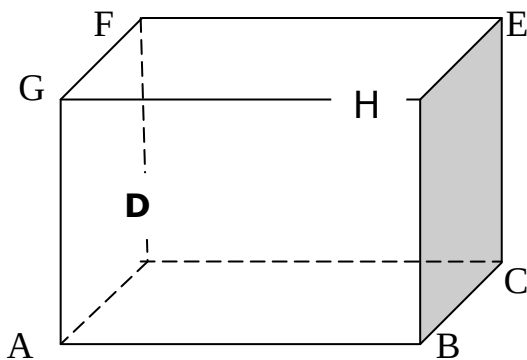
24. In the figure, ABCD is a parallelogram E, F, G and H are mid-points of $\overline{AB}$, $\overline{AD}$, $\overline{BC}$, and $\overline{DC}$ respectively. Which of the following is NOT necessarily true?

- A. The lines $\overleftrightarrow{EF} \parallel \overleftrightarrow{GH}$
- B. $\triangle AEF \sim \triangle CDB$
- C. $a(\triangle CGH) = \frac{1}{8}a(ABCD)$
- D. $\triangle ABF \cong \triangle CBH$



25. ABCD is the base of a rectangular prism as shown in the figure below. If the area of ABCD is 12 sq.cm, $m\angle HAB = 60^\circ$, and $FD = 4\sqrt{3}$ cm, then the volume of the prism is equal to

- A. $24\sqrt{3} \text{ cm}^3$
- B. $64\sqrt{3} \text{ cm}^3$
- C. 144 cm^3
- D. $48\sqrt{3} \text{ cm}^3$



PART III: SHORT ANSWER QUESTIONS ITEMS

WRITE THE SIMPLIFIED ANSWER ON THE ANSWER SHEET PROVIDE

1. What is the *exact value* of $\sec 60^\circ + \cot 45^\circ + \csc 30^\circ$? _____
2. What is the measure of the reference angle for an angle of 260° drawn in standard position?

3. A coin and a die are tossed together. What is the probability that a tail and a prime numbers?

4. What is the probability of a certain event? _____
5. If the variance of a population of 20 set of data value is 16 , each set of data is multiplied by 3 then the new standard deviation is _____.

PART IV: WORKOUT QUESTIONS ITEMS

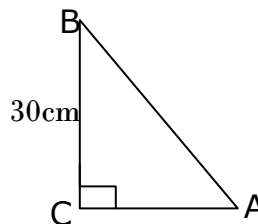
SHOW THE NECESSARILY STEPS CLEARLY

1. In the table given below there is a frequency distribution of a population function V.

V	-2	-1	0	2	4
F	10	10	5	10	5

If an element is drawn randomly from the population then find:

- A. The probability that it is non-negative value?
 - B. The probability that it is non-zero value?
 - C. The probability that it is positive value?
 - D. The probability that it is less than -1?
2. For the right angled triangle given below if $\overline{BC} = 30\text{cm}$ and $m(\angle B) = 30^\circ$, then how long is line segment AB?



3. Given the data 3, 8, 8, 5, 5, 5, 10, 10, 10, and 6. Then find mean, median, mode and range.

===== ***GOOD LUCK*** =====